

Peter Zhang

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Education

STANFORD UNIVERSITY

BS in Computer Science and Mathematics (3.83 | 4.00)

Stanford, CA

May 2027

Experience

Stanford University & Kobilka Lab

Stanford, CA

Primary Researcher (Project Lead)

June 2025 – Now

- Architected a custom molecular design DRL framework, resulting in the discovery of 5 novel lead compounds (U.S. Patents filed) for mitochondrial repair, anti-inflammation, and longevity.
- Achieved a predicted 28,000-fold selectivity for the CB2 receptor, surpassing the widely recognized in vitro benchmark of 1,000+ fold.
- Filtered 1.5M+ compounds via multi-objective rewards, yielded 4 leads with 2000x higher affinity than baseline.

Stanford Vision & Learning Lab with Professor Ron Fedkiw

Stanford, CA

Researcher (LLM & Diff. Modeling)

September 2024 - Now

- Designed a differentiable hair-field estimator integrating volumetric and strand-density priors, reducing calculation time by 60%+ across test subjects.
- Implemented end-to-end distributed PyTorch pipelines with batched augmentations and GPU fine-tuning.
- Applied transformer-based feature encoding for neural rendering, improving generalization under occlusion and pose variation.

Insilico Medicine (HK:3696)

Shanghai, China

Research Intern

July -August 2024

- Engineered AI agents to automate high-volume, repetitive workflows within the drug discovery pipeline, reducing manual intervention by 30 hours/week.
- Optimized drug development efficiency by deploying agentic workflows that handled lead optimization or ADMET prediction, accelerating the research phase.
- Collaborated with cross-functional teams of biologists and data scientists to translate complex pharmaceutical requirements into scalable machine learning solutions.

Reachly-AI (Academia Research Helper)

Stanford, CA

Founder

November 2025 - Now

- Implemented LLM-based systems for profile analysis, powering an academic outreach platform for students
- Built scalable backend workflows for document parsing, professor relevance scoring, and email generation, improving outreach efficiency by saving 20+ hours per week.

Leadership and Activity

- **Immerse the Bay (XR Hackathon)** Grand Prize + UltraLeap + XR Trailblazer Winner (1 out of 300).
- Sixth author, *Realizing optimal radiative cooling walls in building-energy nexus via asymmetric emissivity*, Nexus (Cell Press).
- **USA Swimming Individual Medley Xtreme (IMX) Platinum** | 5th place **USA National Age Group Award**.

Skills and Personal Portfolio

Portfolio: <https://peterzhang04.github.io/>

Languages: Python, C++, Swift UI, Scheme, Java | Native in both English and Mandarin Chinese.

Technical focus: LLM Fine-Tuning, Neural Rendering, Multimodal ML, Data Pipeline Design.